# Script-Controlled ACL – Restrict Record Access Based on Field Value

## Phase 2.4-Technology Stack

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| **Date** | 30 September 2026 |
| **Team ID** | 6ab3ed08a238dc999eb3fda5 |
| **Project Name** | Script-Controlled ACL – Restrict Record Access Based on Field Value |
| **Maximum Marks** | 2 Marks |

## Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a briefjustification for your choice.

## Technology Stack Details

| S.No | Architecture Component / Layer | Technology Chosen | Justification / Purpose |
|---|---|---|---|
| **1** | **Frontend / Client-Side** | React.js with HTML5, CSS3 and Tailwind CSS | Component-based UI that is reusable and fast to update; responsive design works on mobile and desktop; accessible components support WCAG 2.1 AA. |
| **2** | **Backend / Server-Side** | Node.js with Express.js (REST APIs) | Fast, non-blocking processing suits many concurrent learners; JavaScript across frontend and backend simplifies development; large library ecosystem. |
| **3** | **Database / Data Storage** | PostgreSQL<br>Amazon S3 (files) | Strong relational mapping for users, courses, enrollments and credentials, with reliable transactions for payments and certificates. S3 stores course material and certificate PDFs. |
| **4** | **Cloud / Hosting / Deployment** | AWS (EC2 / ECS, RDS)<br>Docker | Easy scalability to meet the 10,000 concurrent user target and high availability; Docker gives identical environments from development to production. |

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|---|---|---|---|
| **5** | **Version Control & CI/CD** | GitHub<br>GitHub Actions | Seamless team collaboration through branches and pull requests; automated testing and deployment on every code push. |
| **6** | **Third-Party APIs / Other Tools** | Razorpay<br>Google OAuth 2.0<br>LinkedIn Share API<br>Twilio (SMS) / Amazon SES (email) | Razorpay enables secure course payments; Google OAuth gives one-click sign-in; LinkedIn lets learners share verified credentials; Twilio and SES send OTPs, alerts and notifications. |